AI-Assisted Fetal Ultrasound Plane Classification for Acquisition Support in Low-Resource Settings: Data Preprocessing Pipeline

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I. Introduction

This project develops a classifier that assigns a fetal ultrasound image to one of four standard planes (abdomen, brain, femur or thorax), so that trained nurses and midwives in low-resource settings receive immediate feedback during acquisition. The model will be pretrained on FETAL_PLANES_DB, which contains 12,400 images from two hospitals in Barcelona [1], [2], and fine-tuned on the African maternal-fetal planes dataset, which contains 450 images from Algeria, Egypt, Ghana, Malawi and Uganda [3], [4]. The two datasets were acquired with different scanners and settings, so their image quality and appearance differ, and models trained in one setting are known to transfer poorly to the other [3].

An initial audit of both datasets defined a preprocessing strategy. This report describes its implementation, the visual diagnostics produced along the way, and the points at which the data forced the plan to change. Section II summarises the audit findings and the missingness analysis, Section III describes each stage of the pipeline with its supporting statistics, Section IV presents the decision log and the rationale for each deviation, and Section V verifies the output of the pipeline. The code, notebook and figures are available at https://github.com/odunayoakinlade/fetal-ultrasound-plane-classification.

II. Data, Missingness and Class Balance

The African metadata has 450 rows and five columns, and the Spanish metadata has 12,400 rows and seven columns, of which 6,561 images from 1,158 patients belong to the four target planes. Because patient numbers restart in each country, patients in the African data are identified by country and patient number together, which gives 127 patients.

Neither file contains an empty cell, so the overall missingness rate is 0% and every row is complete. Missingness is nevertheless present in two less obvious forms. First, not every country collected every plane: Ghana and Uganda contributed no thorax images (Fig. 1). Second, the Spanish data records unknown values as the category "Other", which affects Operator in 19.7% and US_Machine in 3.7% of the four-plane images. Little's MCAR test requires empty cells, so I tested the unknown values directly with chi-square tests and Cramér's V. Whether the operator is unknown depends strongly on the machine ($\chi^2 = 3543.9$, df = 3, $p < 0.001$, $V = 0.73$) and weakly on the plane ($V = 0.17$), and an unknown machine is also linked to the plane ($\chi^2 = 27.2$, $p < 0.001$, $V = 0.06$).

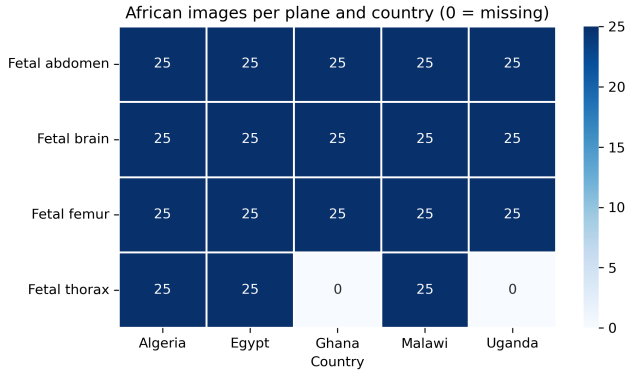


Fig. 1. Images per plane and country in the African dataset. The two empty cells show that Ghana and Uganda contributed no thorax images, the most consequential form of missing data in this project.

These results reject missingness completely at random and support the missing-at-random diagnosis made in the audit. Because Operator and US_Machine are used only to describe the data and to check confounding, never as model inputs, I did not impute them and kept "Other" as its own category.

The missing thorax images make thorax the smallest African class, with an imbalance ratio of 1.67:1, while the Spanish data is dominated by the brain class with a ratio of 4.35:1 (Fig. 2). Each African country also produced images of a single, unique size, and the sites differ in brightness, overlays and screen layout. For instance, Malawi images include a logo, Algerian images contain corner text, and Ghanaian and Ugandan images show caliper marks. These signals reveal where an image was acquired rather than what it shows, and because two countries have no thorax images, a model could learn that images resembling Ghana or Uganda are never thorax. This acquisition shortcut is a form of leakage from the data-generating process [5], and it was the main risk the pipeline had to address.

III. Pipeline Implementation

The pipeline is implemented in a notebook that imports its preprocessing functions and scikit-learn transformers from the project's source modules. Throughout, the development set (80% of the African data) plays the role of the training split for every fitted component, and the Spanish training set (85% of the Spanish data) is used for pretraining.

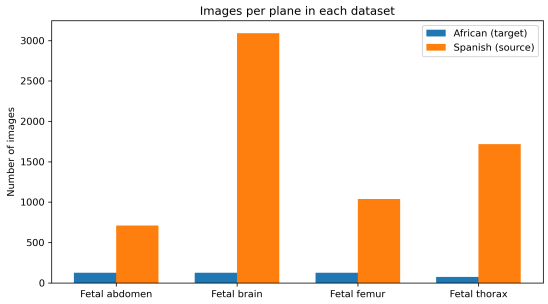


Fig. 2. Images per plane in each dataset before cleaning.

A. Duplicate Removal and Label Encoding

I computed an MD5 hash of every file to find identical copies and a 64-bit perceptual hash of every image to find near-copies, treating two images as duplicates when their hashes differ in at most four bits. In the African data, 94 images shared their MD5 with another file, and 54 pairs fell within the threshold, all with matching labels; removing the second image of each pair reduced the data from 450 to 396 images from 116 patients. The Spanish data contained 591 pairs, four of which carried conflicting labels. I dropped both images of each conflicting pair, because neither label can be trusted without expert review, and one copy of every other pair, leaving 6,059 images from 1,155 patients. No African image appeared in the Spanish data. The planes were then encoded as integers (abdomen 0, brain 1, femur 2, thorax 3), which serve only as class identifiers.

B. Patient-Level Split

Both datasets were split with StratifiedGroupKFold, grouping by patient so that no patient appears in more than one partition, and stratifying by plane. Taking one of five folds as the test set divided the African data into 321 development and 75 test images (81.1% and 18.9%), and taking one of seven folds as the validation set divided the Spanish data into 5,220 training and 839 validation images (86.2% and 13.8%). Every country is represented in the African test set, with between 12 and 18 images each, and the test set is reserved for the final evaluation.

C. Outliers

I defined an outlier as an image that is unusually bright or dark for its own site, measured as the mean pixel intensity inside the ultrasound fan so that the black background does not affect it. This measure is right-skewed (skewness 0.72, excess kurtosis 0.31), and within countries its skewness ranges from 0.22 in Algeria to 1.20 in Uganda, which rules out the Z-score rule because it assumes a normal distribution. I therefore applied the 1.5 × IQR rule within each country, with bounds learned from the development set only by a fitted flagging transformer (Table I).

Nine of the 321 development images (2.8%) were flagged, all above their upper bound (Fig. 3). Four come from a single Egyptian patient, which points to a high brightness setting for that session rather than faulty images, and five are abdomen scans, which are naturally brighter inside the fan. Visual inspection showed that the anatomy remained visible in every flagged image, so all were kept, preserving the full development set of 321 images; deleting them would have removed abdomen images disproportionately and discarded the kind of over-exposed scans the tool will meet in practice. Extreme values are instead capped at the pixel level within each image (Section III-E).

TABLE I
WITHIN-COUNTRY IQR BOUNDS ON FAN BRIGHTNESS (DEVELOPMENT SET)

Country	Q1	Q3	IQR	Lower	Upper
Algeria	50.2	68.4	18.3	22.8	95.8
Egypt	77.3	103.9	26.7	37.2	144.0
Ghana	28.1	39.0	11.0	11.6	55.5
Malawi	60.8	87.2	26.5	21.1	126.9
Uganda	42.8	56.9	14.1	21.7	78.1

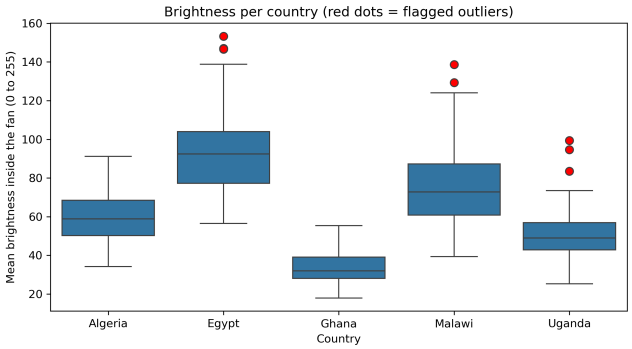


Fig. 3. Fan brightness per country in the development set. Whiskers follow the 1.5 × IQR rule, so the red points are the nine flagged images.

D. Measurements, Collinearity and Association

Because the model learns from pixels, pixel-level correlation is not meaningful, so I described each development image with eight measurements: brightness, contrast, entropy and sharpness inside the fan, the share of the image covered by the fan, and the width, height and aspect ratio of the image. Several measurements are skewed, and the relationship between the most strongly correlated pair is monotonic but visibly curved rather than linear (Fig. 5), so I used Spearman rather than Pearson correlation (Fig. 4).

Following the planned protocol, whenever two measurements had an absolute correlation of 0.85 or more I dropped the one with lower mutual information with the plane (Fig. 6), and I then removed measurements until every variance inflation factor (VIF) was at most 10. Entropy (mutual information 0.072) was correlated with both brightness ($|\rho| = 0.89$, mutual information 0.058) and contrast ($|\rho| = 0.88$, 0.041), so brightness and contrast were dropped. Width, height and aspect ratio take only five distinct combinations and initially had VIFs of 506.0, 250.5 and 503.8; width, the least informative of the three, was removed (VIF = 478.9 at that step), after

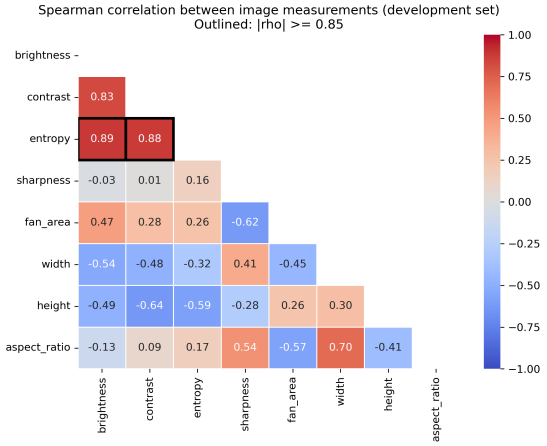


Fig. 4. Spearman correlation between the eight image measurements in the development set. Outlined cells exceed the 0.85 threshold.

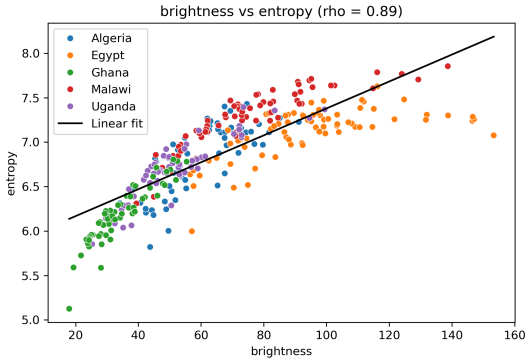


Fig. 5. The most strongly correlated pair, brightness and entropy, coloured by country, with a linear fit. The points curve away from the line at high brightness.

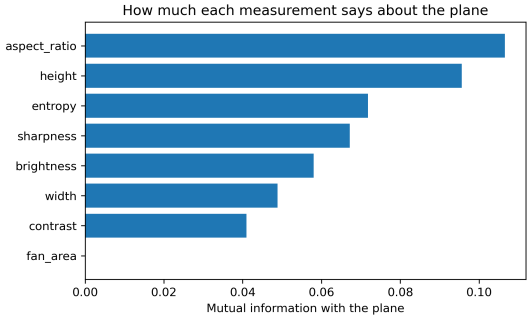


Fig. 6. Mutual information between each measurement and the plane. The two highest-ranked measurements describe image format rather than anatomy.

which the five remaining measurements all had a VIF below 2.62. Both rules are implemented as a transformer that learns which columns to keep, placed with a standard scaler inside a ColumnTransformer and fitted on the development set.

The mutual information ranking also gave independent evidence of the acquisition shortcut. Aspect ratio (0.106) and height (0.096) are the two measurements most informative about the plane (Fig. 6), although they describe the image

TABLE II
ASSOCIATION BETWEEN SITE VARIABLES AND THE PLANE

Variable (partition)	χ^2	df	p	V
Country (African dev.)	35.6	12	< 0.001	0.19
Machine (Spanish train.)	138.3	9	< 0.001	0.09
Operator (Spanish train.)	246.1	9	< 0.001	0.13

format and not the anatomy; they carry information only because image size is fixed per country and two countries have no thorax images. To quantify how strongly site is tied to the plane, I computed chi-square tests and Cramér's V (Table II). All three associations are significant but weak, which indicates that the confounding lies less in which planes each site collected than in how each site's images look.

E. Image Standardisation

Every image passes through the same five steps, chained as a scikit-learn Pipeline: conversion to grayscale, a fan mask that sets everything outside the scan area to black, cropping to the fan's bounding box, clipping of pixel values inside the fan to that image's own 1st and 99th percentiles, and padding to a square followed by resizing to 224 × 224 (Fig. 7). The percentile clipping is a pixel-level form of winsorisation that caps over-exposed regions and the brightest marks without removing any image. When images are passed to the model they are scaled to [0, 1], copied into three channels and standardised with the fixed ImageNet constants expected by the pretrained backbone. None of these steps learns from the data, so the same pipeline can be applied to every partition without leakage.

To see what these steps do to the pixel values, I compared 50 development images at each stage (Fig. 8). Trimming barely changed the average pixel value (63.9 to 65.4) or its spread (standard deviation 48.4 to 48.3) but made the distribution slightly less lopsided (skewness 1.19 to 1.08), which shows that only the most extreme pixels were affected. The scaling steps change only the units, not the shape, so skewness stays at 1.08: dividing by 255 places every value between 0 and 1, and the ImageNet constants rescale the values to match the data the pretrained network was trained on. The result is centred near -1 rather than 0 because these constants come from ImageNet, not from this data. I did not apply a log transform, because scanners already compress ultrasound signals logarithmically.

F. Embeddings and Principal Component Analysis

To describe the processed images in a compact form, I extracted 512-dimensional embeddings from an ImageNet-pretrained ResNet18 [6] for the African images and fitted a Pipeline of a standard scaler and principal component analysis on the development set only. The first component has an eigenvalue of 56.5 and explains 11.0% of the variance, and the first five together explain 33.8%. Ninety-six components retain 90% of the variance and 139 retain 95% (Fig. 9), so the information is spread across many directions rather than concentrated in a few. Because the classifier will be fine-tuned

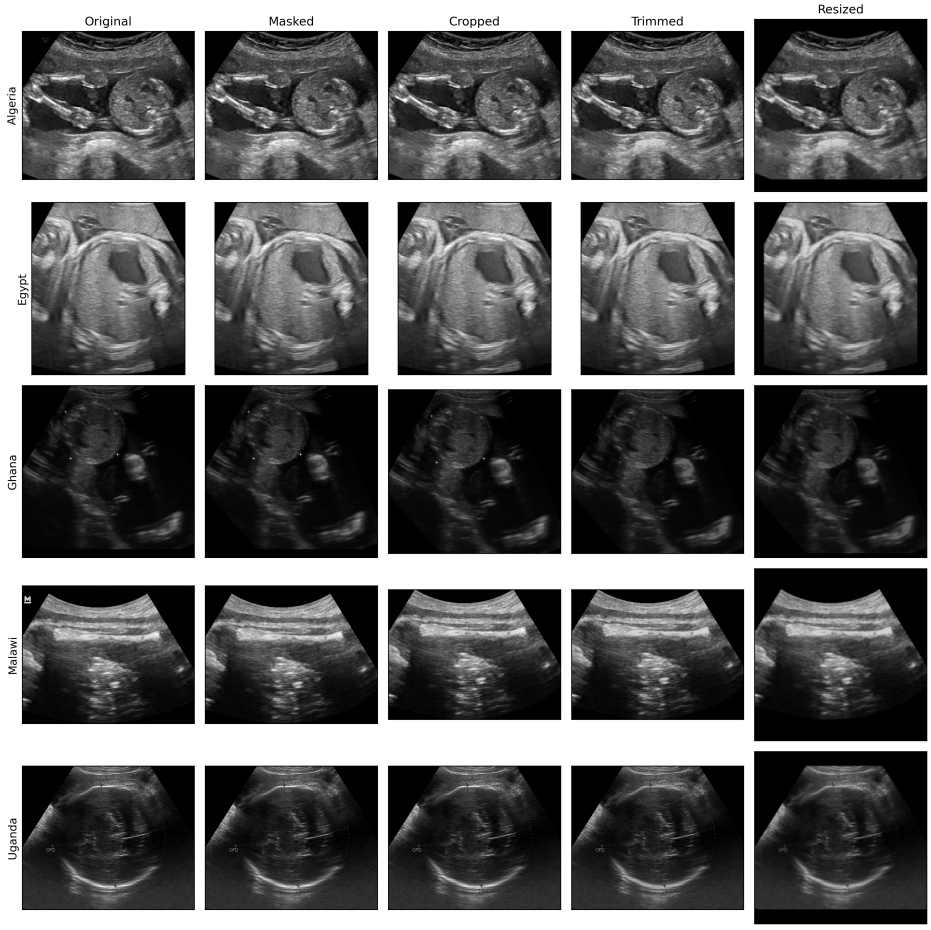


Fig. 7. One image per country at each stage of the image pipeline. Masking removes the Malawi logo and the Algerian corner text, cropping removes the empty border, and padding preserves the aspect ratio of the fan. Caliper marks inside the fan (Ghana, Uganda) remain.

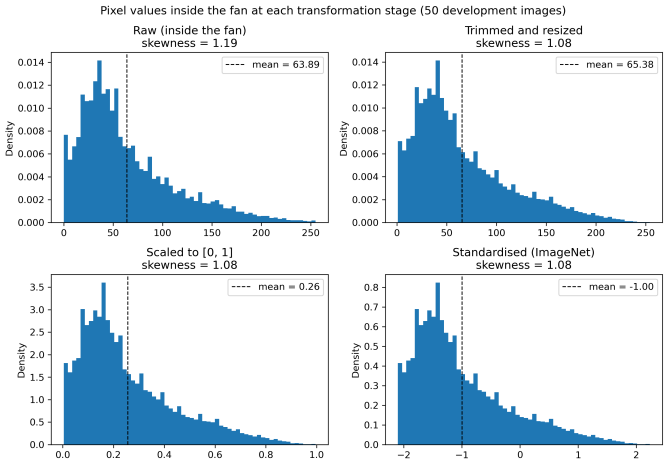


Fig. 8. Distribution of pixel values inside the fan at each transformation stage (50 development images), with means and skewness.

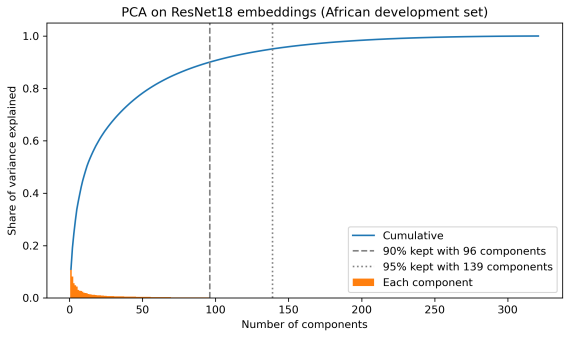


Fig. 9. Variance explained by each principal component of the ResNet18 embeddings (bars) and cumulatively (line), with the 90% and 95% cutoffs (African development set).

end to end, PCA serves here as a diagnostic of embedding redundancy rather than as a model input.

G. Shortcut Test, Class Weights and Augmentation

I repeated the audit's shortcut test on the development set: every image was reduced to a 32 × 32 thumbnail, and a logistic regression, evaluated with patient-grouped five-fold cross-validation and with its scaler refitted inside each fold, predicted the country and the plane. Standardisation did not

TABLE III
DECISION LOG: PLANNED STRATEGY AGAINST IMPLEMENTATION

Step	Planned strategy (Phase 1)	Executed (Phase 2)	Justification metric and figure
Patient key and scope	Country + patient number; source restricted to four planes	As planned	127 African patients; 6,561 source images, 1,158 patients
Duplicates	MD5 and perceptual hash, distance ≤ 4; keep one copy; drop conflicting groups	Pairwise removal (D1)	54 African pairs (450 $\rightarrow$ 396); 591 Spanish pairs, 4 conflicting (6,561 $\rightarrow$ 6,059); 0 cross-dataset matches
Missingness	No imputation; MAR for "Other"; thorax gap handled in evaluation	As planned; figure redesigned (D2)	0% empty cells; Operator unknown vs machine $V = 0.73$, $p < 0.001$ (Fig. 1)
Label encoding	Integers 0–3	As planned	Four classes; thorax 61 of 396 after cleaning (Fig. 2)
Split	StratifiedGroupKFold by plane and patient: 80/20 target, 85/15 source	Patient-level shares (D3)	321/75 and 5,220/839 images; 0 patients and 0 duplicate pairs across partitions
Outlier detection	IQR rule within site on mean intensity; bounds from training data	Measured inside the fan (D4)	Skewness 0.72, kurtosis 0.31; 9 of 321 flagged (2.8%) (Fig. 3, Table I)
Outlier treatment	No deletion; clip each image to its 1st–99th percentiles	As planned	Development set kept at 321; skewness 1.19 $\rightarrow$ 1.08 (Fig. 8)
Association	Chi-square and Cramér's V; mutual information	As planned	Cramér's V 0.09–0.19, all $p < 0.001$ (Table II, Fig. 6)
Collinearity	Spearman $\lvert\rho\rvert \geq 0.85$, VIF > 10; keep higher mutual information	Different features dropped (D5)	Brightness ($\rho = 0.89$), contrast (0.88), width (VIF 478.9) dropped; max VIF 506.0 $\rightarrow$ 2.62 (Figs. 4–6)
Fan mask	Largest bright connected region	Convex hull added (D6)	Anatomy restored in dark Ghanaian and Ugandan images (Fig. 7)
Scaling	Crop, pad and resize to 224; [0, 1]; fixed backbone constants; no log transform	As planned	Mean 0.26, SD 0.19 after [0, 1]; mean -1.00, SD 0.83 after ImageNet scaling (Fig. 8)
PCA	On standardised backbone embeddings, fitted on training data	African images only (D7)	96 components for 90%, 139 for 95%; PC1 11.0% (Fig. 9)
Shortcut check	Repeat after preprocessing	Shortcut persists (D8)	Country accuracy 91.5% $\rightarrow$ 93.1% vs plane 54.2% $\rightarrow$ 51.8% (Table IV)
Class imbalance	Weighted loss from training folds; no resampling	As planned	Imbalance 1.67:1 and 4.35:1; thorax weight 1.54 (Fig. 2)
Augmentation	Training only, incl. quality degradation; fixed in Phase 2	Specified	Brightness, contrast, gamma, blur, noise, rotation, flip
Model evaluation	Leave-one-domain-out, overlay invariance, macro-F1, calibration	Deferred (D9)	Requires a trained model

TABLE IV
SHORTCUT TEST ACCURACY ON 32×32 THUMBNAILS (%)

	Country	Plane
Before standardisation	91.5	54.2
After standardisation	93.1	51.8
Baseline (most common class)	24.0	28.0

reduce the shortcut (Table IV). Masking removes overlays outside the fan, but each scanner produces a fan with a characteristic outline, which fills the frame once the image is cropped (Fig. 7), and brightness and texture still differ between sites (Fig. 3).

To address the class imbalance without repeating images, I computed balanced class weights for a weighted cross-entropy loss [7] from the training partitions only, giving thorax a weight of 1.54 in the African development set and weights between 0.51 (brain) and 2.21 (abdomen) in the Spanish training set. I also fixed the augmentation that will be applied to training batches only: brightness and contrast changes of up to $\pm 20\%$, gamma between 0.8 and 1.2, Gaussian blur and speckle noise to imitate lower-quality equipment, rotations of up to $\pm 10^\circ$ and left–right flips. Vertical flips are excluded because the probe is always at the top of the image.

IV. DECISION LOG AND DEVIATIONS

Table III compares each item of the initial strategy with its implementation, together with the metric and figure that justify it. Items marked D1 to D9 deviated from the plan and are explained below.

D1. Pairwise duplicate removal. I dropped the second image of each pair within the threshold rather than collapsing chains of near-copies into one group; this applies the threshold exactly and removed 502 Spanish images, close to the roughly 500 expected.

D2. Missingness figure. A missingness matrix would be blank because no cell is empty, so Fig. 1 shows the missing plane–country combinations instead, and chi-square tests quantify the implicit "Other" values.

D3. Split shares. Keeping whole patients together made the shares 81/19 and 86/14 rather than exactly 80/20 and 85/15.

D4. Outlier measure. Measuring brightness inside the fan rather than over the whole image, so that the background does

TABLE V
MATRIX DIMENSIONS ($N \times P$) BEFORE AND AFTER THE PIPELINE

Matrix	Before	After
African images	450 × 180,000–300,000 px	396 × 50,176
Spanish images	6,561 × up to 1.18 M px	6,059 × 50,176
Model input (African)	—	396 × 3 × 224 × 224
Measurements (dev.)	321 × 8	321 × 5
Embeddings (dev.)	321 × 512	321 × 139

not distort it, flagged nine images instead of the seven flagged in the audit; none was deleted.

D5. Dropped measurements. The strongest correlations were between brightness, contrast and entropy rather than between the size measurements, as I had expected, so brightness, contrast and width were dropped (Fig. 4).

D6. Convex-hull fan mask. The planned mask cut anatomy out of dark Ghanaian and Ugandan scans, the two sites without thorax images; taking the convex hull of the largest region restores the full fan while logos and text stay removed (Fig. 7).

D7. PCA scope. The embeddings and PCA cover the African images only, which describe the target domain and keep the computation feasible on a laptop CPU.

D8. Persistent shortcut. Standardisation did not reduce how well the country can be predicted (Table IV). Augmentation and leave-one-domain-out evaluation will address and measure this during training, and a common fan-shaped mask applied to every image is a candidate fix.

D9. Model-dependent evaluation. Leave-one-domain-out evaluation, the overlay invariance check, macro-F1, per-class recall and calibration require a trained model and are deferred to the modelling phase.

V. PIPELINE OUTPUT VERIFICATION

Table V gives the dimensions of each matrix before and after the pipeline. The reductions remove only redundant or non-anatomical information. The 54 African and 502 Spanish images removed as duplicates add no independent information, and the eight conflicting images could not be labelled reliably. Each standardised image keeps the full fan and discards only the border, the overlays outside it and empty space, and trimming changed the mean and standard deviation of pixel values inside the fan by less than 2.5% (Fig. 8). The three measurements dropped were each strongly correlated with one that was kept, so the information they carried is retained, and the 139 principal components retain 95% of the variance of the embeddings.

A final set of twelve automated checks passed: no patient and no near-duplicate pair crosses partitions in either dataset, no African image appears in the Spanish data, every processed image is 224 × 224 and non-blank with one image per cleaned record, the model input has shape 3 × 224 × 224, and the outlier bounds, the measurement ColumnTransformer and the embedding Pipeline were each fitted on exactly the 321 development images.

VI. LIMITATIONS AND NEXT STEPS

The main limitation is the acquisition shortcut, which preprocessing alone did not remove. Caliper marks and measurement labels inside the fan cannot be masked without damaging anatomy, although deployed images will rarely contain them because planes are identified before measurements are taken. The outlier check and the embeddings cover the African images only, and the missing thorax images in Ghana and Uganda cannot be recovered, so thorax performance will be reported only for the three countries that collected it. The next phase will pretrain and fine-tune the classifier on the processed data, apply the specified augmentation and class weights, and evaluate it on the held-out test set and with leave-one-domain-out evaluation, which will show whether the shortcut persists in a trained model.

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